



Pre-positioning of emergency supplies for disaster response

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ABSTRACT

Pre-positioning of emergency supplies is one mechanism of increasing preparedness for natural disasters. The goal of this research is to develop an emergency response planning tool that determines the location and quantities of various types of emergency supplies to be pre-positioned, under uncertainty about if, or where, a natural disaster will occur. The paper presents a two-stage stochastic mixed integer program (SMIP) that provides an emergency response pre-positioning strategy for hurricanes or other disaster threats. The SMIP is a robust model that considers uncertainty in demand for the stocked supplies as well as uncertainty regarding transportation network availability after an event. Due to the computational complexity of the problem, a heuristic algorithm referred to as the Lagrangian L-shaped method (LLSM) is developed to solve large-scale instances of the problem. A case study focused on hurricane threat in the Gulf Coast area of the US illustrates application of the model.

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1. Introduction

Extreme events such as tornados, earthquakes or hurricanes can strike a community with little warning and leave much devastation and many casualties behind. The main goal of emergency response efforts is to provide shelter and assistance to disaster victims as soon as possible. In order to achieve this goal, several types of supplies are essential, and one element of preparedness planning can be pre-positioning of supplies at strategic locations so that they are readily available when needed. An obvious difficulty in creating an effective pre-positioning plan is uncertainty about whether or not natural disasters will occur, and if they do, where and with what magnitude.

Much of the effort in emergency response research focuses on disaster management following events. Primary concerns are often the integration of information pertaining to the location and capacities of resource providers, the spatial distribution of the victims, immediate requirements for search-and-rescue, etc. (Al-qurashi, 2004; Fiedrich et al., 2000; Parentela and Nambisan, 2000; Sherali et al., 2004; Tsai et al., 2002). Houming et al. (2008) and Sheu (2007) describe post-event distribution of emergency supplies through a three-layer supply chain that connects relief suppliers, distribution centers and victims. Haghani and Oh (1996) developed a multi-commodity multimode network flow model to help emergency response managers organize detailed load plans for moving commodities after an event. Barbarosoglu and Arda (2004) extended that work to consider uncertainties in available supplies, demands and network arc capacities through definition of a set of scenarios. The inclusion of uncertainties is an important advance in the analysis, but the focus is still post-event response. Related work has been done by Ozdamar et al. (2004), who created a dynamic model to generate multi-period vehicle routes and schedules along with commodity load-unload assignments.

The location and capacities of the resource providers are key components in managing response efforts after an event, but relatively little research has been conducted on the topic of a priori planning for pre-positioning specific resources. Military

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forces use pre-positioning strategies for equipment, ammunition, etc. to facilitate rapid and effective response to conflicts (Johnstone et al., 2004). However, limited planning tools resulting from formal modeling techniques (optimization or simulation) have been developed, and the existing models generally do not account for uncertainty in demands or damage in the network (e.g., roads and facilities that may be obstructed, destroyed or unavailable).

There has been some work on locating first responders for incidents. For example, Saccomanno and Allen (1988) used a location model to determine sites for response-capable units (e.g., fire companies or police units) that could provide aid in case of spills of dangerous goods on a rural road network. Sathe and Miller-Hooks (2005) studied the relocation of first-response units (e.g., military units, police forces) in order to maintain protection coverage to critical facilities under disaster conditions. However, locating first-response units is different from locating and sizing stocks of supplies, where multiple commodities must be considered, the commodities may have differing storage requirements and transportation costs, etc.

In the fall of 2005, hurricanes Katrina, Wilma and Rita caused more than \$100 billion in damage, and highlighted the inadequacy of existing preparedness strategies. Since then, some research effort has been aimed at devising pre-positioning plans for emergency supplies. Ukkusuri and Yushimoto (2008) develop a model to choose locations for pre-positioned supplies in a way that maximizes the probability that demand points can be reached from at least one supply facility under conditions where facility locations and links in the transportation network may become unusable. Balci and Beamon (2008) develop a maximal covering location model that determines the number and location of fixed-capacity distribution centers in a relief network, and the amount of relief supplies to be stocked at each center. The model considers pre- and post-disaster budget restrictions but does not consider network reliability. Demands are specified through the use of scenarios, but there is an implicit assumption that in each scenario there is demand in only one location.

In this paper, we formulate a model that combines facility location, decisions on stocking levels for emergency supplies, and distribution of those supplies to multiple demand locations after an event, with uncertainty about demand, survival of pre-positioned stocks and the condition of the transportation network following an event. Specifically, we consider the possibility that locations that stock supplies may be destroyed in an event, causing loss (partial or complete) of whatever is stored there. We also consider the possibility that parts of the transportation network used to move supplies may become unusable after an event. The decisions considered in the model include the locations and sizes of storage facilities, the amounts of various materials stocked in each facility, and the distribution of those supplies to one or more locations in response to a variety of potential events.

The model formulated is a two-stage stochastic mixed integer program (SMIP). Stochastic programming is a general purpose technique that deals with uncertainty in the input values to optimization models. Nonanticipative first-stage decisions are made in the presence of uncertainty about future realizations (in this case hurricane effects on the system including demand variations and network damages). Second-stage (recourse) decisions are made after the realization of the random problem elements is known, and are conditional on the first-stage decisions. The objective is to make the optimal set of first-stage decisions, under uncertainty about the conditions to be faced in the second stage (Birge and Louveaux, 1997).

The model uses a set of discrete scenarios (with probabilities) to represent potential locations and magnitudes of the uncertain disaster events. The first-stage variables in the stochastic optimization include the storage facility locations and sizes, as well as stocking decisions for various types of supplies. The second-stage (recourse) decisions involve the distribution of available supplies in response to specific scenario events and network conditions. The structure of this problem formulation lends itself to a specialized solution method that allows instances of relatively large scale to be solved efficiently.

In Section 2, we describe the formulation and set the stage for discussion of the specialized solution method, which is described in Section 3. In Section 4, we illustrate the use of the model through a case study for pre-positioning of supplies in anticipation of hurricanes in the southeastern US. Section 5 evaluates the effectiveness of the specialized solution algorithm as compared to commercial mixed-integer optimization software, and Section 6 provides conclusions and directions for further work.

2. Mathematical formulation

We consider a set of commodities that may be pre-positioned in storage facilities and for which there is likely to be demand after an event like a hurricane. In the example application in Section 4, we consider three commodities (prepackaged meals, water and medical kits), but in general the list of commodities might also include blankets, cots, ice, clothing, etc. The set of commodities is denoted by K and indexed by $k \in K$. In some cases, it may be desirable to “bundle” individual commodities to make sure they are provided together (e.g., beds and blankets, food and water, etc.). A bundled commodity can be defined that includes specific elements in fixed proportion. This is still handled by the algorithm as a single commodity, and does not change the structure of the formulation, but decisions to acquire, stock and transport units of this bundled commodity have costs, storage requirements, etc., that reflect the aggregation.

In the aftermath of an event, there will be demands for these commodities at specific locations in a set I , indexed by $i \in I$. At the planning stage, the demand for commodity k at location i , v_i^k , is uncertain because it is not yet known if, or where, an event will occur. This uncertainty is a critical part of the model. In any specific location, there may be a relatively large probability of no demand over the time horizon for which planning is being done. What constitutes a location, i , may also vary across specific applications of the model. In the example in Section 4, we adopt a regional perspective (10 states in the

southeastern US) and the locations specified are cities. However, for more detailed planning at a smaller geographic scale, the locations might be designated shelters within a metropolitan area, for example.

Uncertainty is modeled through the use of a set S of discrete scenarios indexed by $s \in S$, each with a probability of occurrence, P_s . The definition of a scenario includes the forecasted demand by commodity and location, v_i^{ks} . An illustration of the construction of a set of scenarios for potential hurricane events is given in Section 4.

Supplies can be pre-positioned at a location i if a storage facility is made available there. For costing purposes, we define facilities to be in one of a discrete set, L , of size categories, indexed by $l \in L$. The overall capacity (e.g., square feet of available space) of a facility in category l is M_l , and choosing to open a facility of size category l in location i incurs a fixed cost, F_{il} . Let y_{il} be a binary decision variable equal to 1 if there is a supply facility of capacity category l located at node i , and 0 otherwise. This is one of the primary sets of decisions in the model.

If a facility is made available at location i , various commodities can be stocked there, subject to the capacity limits of the facility. Let b^k be the unit space requirement for commodity k , and r_i^k be the amount of commodity k pre-positioned at location i . The r_i^k values are the second major set of first-stage decision variables in the model. A decision to stock a particular commodity results in a unit acquisition cost, q^k . Material of type k that is not used in scenario s , denoted z_i^{ks} , incurs an additional unit holding cost, h^k (accounting for general inventory-related or spoilage costs). On the other hand, if demand for a particular commodity cannot be met, we denote w_i^{ks} as the shortage of commodity k at location i in scenario s , and associate a unit penalty cost, p^k , with that shortage.

In specific scenarios, some or all of the materials pre-positioned at a given location may be destroyed by the event. This is a risk incorporated into the analysis. We denote by ρ_i^{ks} the proportion of stocked material of commodity k at location i that remains usable in scenario s ($0 \leq \rho_i^{ks} \leq 1$). Specification of the ρ_i^{ks} values becomes part of the scenario definitions.

The facility and stocking decisions are made before knowledge of any event scenario is available. After an event occurs, the surviving stocks of the various commodities are distributed across a transportation network to meet demands. Let c_{ij}^{ks} be the unit cost of transporting commodity k across link (i, j) in the transportation network and let x_{ij}^{ks} be the amount of commodity k shipped across the link in scenario s . The definition of scenarios may also make some transportation links unusable (or only partially usable). In general, we will denote by U_{ij}^s the available capacity on link (i, j) in the transportation network in scenario s , with the understanding that U_{ij}^s may be zero in some scenarios. Because the “consumption” of link capacity by units of different commodities may not be the same, we use the coefficient u^k to relate flow of commodity k to capacity of transportation links.

To reflect the connection between the facility location and demand elements of the problem and the transportation elements, we will assume that the demand locations and potential facility locations are at nodes in the transportation network. In general, the set of locations of interest may be a subset of all the nodes in the network, and we will denote the full set of nodes as N . The transportation links (i, j) are assumed to be elements of an arc set A , with $i \in N$, $j \in N$.

The model can then be formulated as an optimization problem (P1), in which the objective is to minimize the expected costs over all scenarios resulting from the selection of the pre-positioning locations and facility sizes, the commodity acquisition and stocking decisions, the shipments of the supplies to the demand points, unmet demand penalties and holding costs for unused material.

$$(P1) \quad \min \quad \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{k \in K} \sum_{i \in I} q_i^k r_i^k + \sum_{s \in S} P_s \left[\sum_{(i,j) \in A} \sum_{k \in K} c_{ij}^{ks} x_{ij}^{ks} + \sum_{i \in I} \sum_{k \in K} (h^k z_i^{ks} + p^k w_i^{ks}) \right] \quad (1)$$

Subject to: (i) Flow conservation

$$\sum_{j \neq i \in N} x_{ji}^{ks} + \rho_i^{ks} r_i^k - z_i^{ks} = \sum_{j \neq i \in N} x_{ij}^{ks} + v_i^{ks} - w_i^{ks} \quad \forall i \in N, k \in K, s \in S \quad (2)$$

(ii) Arc capacity

$$\sum_{k \in K} u^k x_{ij}^{ks} \leq U_{ij}^s \quad \forall (i, j) \in A, s \in S \quad (3)$$

(iii) Open facilities and facility capacity

$$\sum_{k \in K} b^k r_i^k \leq \sum_{l \in L} M_l y_{il} \quad \forall i \in I \quad (4)$$

(iv) Number of facilities per node

$$\sum_{l \in L} y_{il} \leq 1 \quad \forall i \in I \quad (5)$$

(v) Non-negativity constraints

$$\begin{aligned} y_{il} &\in (0, 1) & \forall i \in I, l \in L \\ r_i^k &\geq 0 & \forall i \in I, k \in K \\ x_{ij}^{ks} &\geq 0 & \forall (i, j) \in A, k \in K, s \in S \\ z_i^{ks}, w_i^{ks} &\geq 0 & \forall i \in I, k \in K, s \in S \end{aligned} \quad (6)$$

The first constraint represents conservation of flow in the network at each node, for every commodity and every scenario. This conservation of flow uses the undamaged amounts of the stocked commodities at locations ($\rho_i^{ks} r_i^k$) in scenario s and indirectly defines the unmet demand and the unused stocks. The second constraint ensures that the link flow does not surpass the arc capacity, which may change randomly in each scenario. The third constraint makes certain that stocked commodities are assigned to open facilities and that the space taken by these resources does not exceed the facility capacity. The fourth constraint limits the number of open facilities at node i to one.

Problem (P1) is a two-stage stochastic mixed-integer linear program. The y_{il} and r_i^k variables are the first-stage decisions, made before there is knowledge of any specific scenario outcome. The x_{ij}^{ks} , w_i^{ks} and z_i^{ks} variables are scenario-specific and constitute the second-stage, or recourse, variables. For relatively small problem instances, it is reasonable to solve (P1) directly in extensive form (i.e., expanding out the scenario-specific constraints and creating an equivalent deterministic mixed-integer programming (MIP) problem). This can be accomplished with commercial MIP software. However, as the size of the problem instance increases (larger numbers of locations, commodities and/or numbers of scenarios), direct solution using commercial software becomes computationally unattractive. For example, a problem with three possible facility sizes, 10 commodities, 100 scenarios and a network with 200 nodes and 600 arcs would imply a problem with more than 1 million variables (600 of which are binary) and about 200,000 constraints. A problem of this scale might be solvable with a good commercial MIP solver like CPLEX, but would likely require several hours of computing time, at best. To make the model much more usable, there is a significant advantage to developing a special-purpose solution algorithm that will operate much faster and scale better to larger problem instances.

We can recognize in problem (P1) a combination of structural elements. There is a location problem for the storage facilities, a space allocation problem to assign stocks of commodities to the facilities, and a multi-commodity network flow problem for each scenario to distribute the available stocks to specific demands. In what follows, we exploit this overall problem structure to create a specialized solution procedure. The first step in creating this procedure is to define an augmented network that allows the second-stage portion of the problem to be solved efficiently. Then Section 3 of the paper shows how a general approach to stochastic mixed-integer problems (the Integer L-Shaped Method) can be adapted to take advantage of the structure within the first-stage part of the problem and the nature of the connections between the first and second stages.

In the augmented network for the second-stage solution, the first-stage variables (y_{il} and r_i^k) are assumed known. Each location that is a storage facility site ($y_{il} = 1$ for some l) is divided into two nodes, as shown in the small three-node example in Fig. 1. A copy of the network with data specific to each scenario is created. In the scenario s copy, the “demand” node at location i contains the demands (v_i^{ks}) and serves as the connection point to the transportation system. A second node (indicated by the “prime” notation) is the supply, with available supplies, r_i^k . It is connected to its demand counterpart by a link with zero cost and individual commodity capacities, $\rho_i^{ks} r_i^k$. All supply nodes are connected to a sink node by arcs with unit costs, h^k (by commodity), and unlimited capacity. A “super-source” node is created and connected to all demand nodes by arcs with unit commodity flow costs, p^k , and unlimited capacity. The super-source is also connected to the sink node by an arc with zero cost and unlimited capacity. The super-source node has a supply of commodity k equal to $\sum_i v_i^{ks}$. The sink node has a demand for commodity k equal to $\sum_i r_i^k$.

Individual transportation links in the network have unit costs, c_{ij}^k , and overall capacities, U_{ij}^s . This multi-commodity flow problem is further complicated by the fact that different commodities may consume different amounts (u^k) of the limited arc capacities. In a general case, we want to solve this sub-problem directly. However, in many practical cases, the most important aspect of the transportation link capacity constraints may be whether the link is usable or not (i.e., is $U_{ij}^s = 0$ or not). This is similar to the argument made by Ukkusuri and Yushimoto (2008) that focuses on the reliability of arcs in the network. In this special case, the multi-commodity aspect of the network flow problems can be relaxed to a series of single-commodity problems, further simplifying the solution.

For our current purposes, we will be concerned with the special case where links for which $U_{ij}^s = 0$ in a specific scenario are removed, and the remaining transportation links are assumed to have unlimited capacity. In this case, the commodity flows become separable, and the second-stage problems can be solved as a set of $|S| \times |K|$ single-commodity network flow

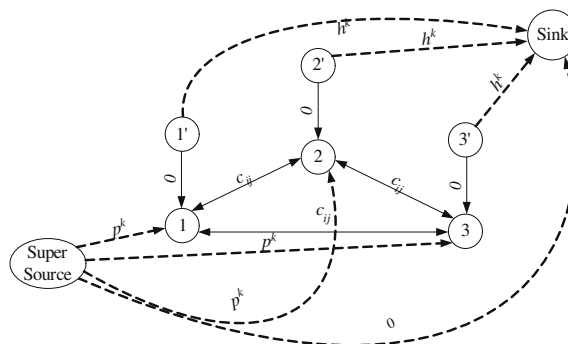


Fig. 1. Network representation, including link costs.

problems. This special case has important practical usefulness, and in a subsequent paper we will explore the algorithm modifications necessary to accommodate the more general case where constraint (3) is incorporated explicitly.

In the network flow solutions, the flows on the “real” network arcs are the x_{ij}^{ks} variables, the flows on the arcs from the super-source to the demand nodes are the w_i^{ks} variables, and the flows on the arcs from the supply nodes to the sink are the z_i^{ks} variables. The flow conservation constraints are enforced by the network structure and the cost of each network flow solution is the minimum cost for commodity k in scenario s , given the first-stage variables, y_{il} and r_i^k , are assumed fixed.

In the special case where the multi-commodity network flow problem decomposes into a series of single-commodity problems, extremely efficient algorithms (Ahuja et al., 1993) can be used for the network flow portion of the overall solution approach.

3. Overall solution methodology

We will define $H(y, r)$ as the expected value of the second-stage sub-problem in (P1), given specific values for the first-stage variables (y_{il} and r_i^k). Then problem (P1) can be re-written as (P2).

$$(P2) \quad \text{Min} \quad \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{i \in I} \sum_{k \in K} q^k r_i^k + H(y, r) \quad (7)$$

$$\text{s.t.} \quad \sum_{k \in K} b^k r_i^k \leq \sum_{l \in L} M_l y_{il} \quad i \in I \quad (8)$$

$$\sum_{l \in L} y_{il} \leq 1 \quad i \in I \quad (9)$$

$$y_{il} \in (0, 1) \quad i \in I, l \in L \quad (10)$$

$$r_i^k \geq 0 \quad i \in I, k \in K \quad (11)$$

In (P2), only the first-stage variables appear explicitly, but to evaluate the objective function for a selection of y_{il} and r_i^k variables, all the scenario-specific second-stage problems have to be solved to compute $H(y, r)$. The L-shaped method, developed by Van Slyke and Wets (1969), reduces the number of second-stage problems that have to be solved by using outer linearization to approximate $H(y, r)$. The expected value of the second-stage problems is a convex function (Birge and Louveaux, 1997), so an approximation based on a series of linear support functions will always lie on or below $H(y, r)$.

Laporte and Louveaux (1993) developed an extension of the L-shaped method to deal with problems that include integer variables in the first stage. This is important for our application because the y_{il} are binary. The general direction of this extension is to create a method for finding feasible solutions for the integer variables and then to use the linear outer approximation to $H(y, r)$ for the continuous variables. In our adaptation of the integer L-shaped method for problem (P2), we construct a means of always maintaining feasibility for the integer variables and focus on the outer linearization in the commodity variables, r_i^k .

The outer linearization is constructed via a set of approximations, each constructed at some specific set of r_i^k values. If the individual linear equations are indexed by t , the general form is $E_t + \sum_{i \in I} \sum_{k \in K} e_{it}^k r_i^k$. Each construction is based on the partial derivatives of $H(y, r)$ for changes in r_i^k , but these values are just the dual variables from the flow conservation constraints in the individual commodity network flow problems. If the dual variable for the flow conservation constraint for commodity k at location i in scenario s (when the specific values of r_i^k are r_{it}^k) is denoted γ_{it}^{ks} , then slope coefficients of the linear approximation are computed as:

$$e_{it}^k = \sum_{s \in S} P_s \gamma_{it}^{ks} \quad (12)$$

and the intercept term is:

$$E_t = \sum_{s \in S} \sum_{i \in I} \sum_{k \in K} P_s \gamma_{it}^{ks} r_{it}^{ks} \quad (13)$$

Problem (P2) is then modified to (P3):

$$(P3) \quad \text{Min} \quad \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{i \in I} \sum_{k \in K} q^k r_i^k + \theta \quad (14)$$

$$\text{s.t.} \quad \sum_{k \in K} b^k r_i^k \leq \sum_{l \in L} M_l y_{il} \quad i \in I \quad (15)$$

$$\sum_{l \in L} y_{il} \leq 1 \quad i \in I \quad (16)$$

$$\theta \geq E_t - \sum_{i \in I} \sum_{k \in K} e_{it}^k r_i^k \quad t = 1, \dots, T \quad (17)$$

$$y_{il} \in (0, 1) \quad i \in I, l \in L \quad (18)$$

$$r_i^k \geq 0 \quad i \in I, k \in K \quad (19)$$

T represents the total number of linear approximations (often referred to as “optimality cuts”) made thus far, at any stage of the overall solution. Each time we solve (P3), we check the solution by solving the network flow problems to obtain an actual value of $H(y, r)$. At each iteration, either $\theta = H(y, r)$ and an optimal solution has been obtained or $\theta < H(y, r)$ and the optimality cuts created thus far are insufficient. In this case, a new optimality cut must be added and the solution of (P3) repeated.

Problem (P3) can be solved effectively using Lagrangian Relaxation. If constraints (15) are relaxed and associated with multipliers μ_i , the relaxed problem can be written as (P4).

$$(P4) \quad \text{Min} \quad L(\mu) = \sum_{i \in I} \sum_{l \in L} F_{il} y_{il} + \sum_{i \in I} \sum_{k \in K} q^k r_i^k + \theta + \sum_{i \in I} \mu_i \left(\sum_{k \in K} b^k r_i^k - \sum_{l \in L} M_l y_{il} \right) \quad (20)$$

$$\text{s.t.} \quad \sum_{l \in L} y_{il} \leq 1 \quad i \in I \quad (21)$$

$$\theta \geq E_t - \sum_{i \in I} \sum_{k \in K} e_{it}^k r_i^k \quad t = 1, \dots, T \quad (22)$$

$$y_{il} \in (0, 1) \quad i \in I, l \in L \quad (23)$$

$$r_i^k \geq 0 \quad i \in I, k \in K \quad (24)$$

For given values of the Lagrange multipliers, μ_i , (P4) clearly separates into two smaller problems, one in the y_{il} variables and the other in the r_i^k variables. These are shown below as sub-problems (P4a) and (P4b).

$$(P4a) \quad \text{Min} \quad \sum_{i \in I} \sum_{l \in L} (F_{il} - \mu_i M_{il}) y_{il} \quad (25)$$

$$\text{s.t.} \quad \sum_{l \in L} y_{il} \leq 1 \quad i \in I \quad (26)$$

$$y_{il} \in (0, 1) \quad i \in I, l \in L \quad (27)$$

$$(P4b) \quad \text{Min} \quad \sum_{i \in I} \sum_{k \in K} (q^k + \mu_i b^k) r_i^k + \theta \quad (28)$$

$$\text{s.t.} \quad \theta \geq E_t - \sum_{i \in I} \sum_{k \in K} e_{it}^k r_i^k \quad t = 1, \dots, T \quad (29)$$

$$r_i^k \geq 0 \quad i \in I, k \in K \quad (30)$$

Sub-problem (P4a) is solved using a greedy algorithm. For each location, i , identify the coefficient $(F_{il} - \mu_i M_{il})$ that has the smallest value. If that coefficient is negative, open a facility of that size category (l) at the location i ($y_{il} = 1$). If the smallest coefficient is 0 or positive, open no facility at i .

Sub-problem (P4b) is a linear program that determines the optimal quantities of different commodities to stock, based on their purchase price and on how much storage space they require, while considering their shipment costs and the penalties incurred when unavailable. The solution for (P4b) is obtained with a commercial linear programming solver. Because the constraints that link the facility location variables, y_{il} , and the commodity variables, r_i^k , have been relaxed, the solutions to (P4a) and (P4b) may not be consistent (i.e., commodities may be stocked at places with no facilities, or in excess of facility capacity). As the correct values of μ_i are approached, this inconsistency disappears.

For any values of the Lagrangian multiplier μ_i , the Lagrangian problem $L(\mu)$ constitutes a lower bound on the optimal objective function value of the original problem (Fisher, 1985). To obtain the lower bound value closest to the optimal solution, one must solve the Lagrangian multiplier problem $L^* = \max_{\mu} L(\mu)$. Thus, the process of solving (P3) is iterative, alternating between solving (P4a) and (P4b) for fixed values of the μ_i , and updating the multipliers (using sub-gradient search). At each Lagrangian iteration, an upper bound on the solution to (P3) is easily obtained by adjusting the y_{il} variables to ensure feasibility in the relaxed constraints. In general, the Lagrangian Relaxation approach cannot be guaranteed to converge to the optimal solution of (P3), so it must be considered a heuristic. However, in this particular application, the computational experience has been very good. This is discussed in further detail in Section 5.

The overall solution procedure is called the Lagrangian L-Shaped Method (LLSM). The following algorithm statement describes the full LLSM algorithm. The L-shaped method is used as the master problem, the Lagrangian relaxation is used to solve the current problem (P3) and a network simplex algorithm solves the minimum cost flow problems finding the re-course function value, $H(y, r)$.

3.1. LLSM procedure

Let ϕ^* be the estimated optimal objective value and denote by ϕ_{P4a} and ϕ_{P4b} the portions of the objective function value associated with solving sub-problems (P4a) and (P4b). Let δ be the current iteration number in the Lagrangian relaxation and T be the number of optimality cuts (of the form shown in Eq. (17)) introduced to approximate $H(y, r)$.

Step 0. Set $T = \delta = 0$; $\mu_i^0 = 0, \forall i \in I$; $\phi^* = \infty$.

Step 1. Set $\delta \leftarrow \delta + 1$. Solve (P4a) and (P4b) with the current values of the Lagrange multipliers, μ_i^δ . Denote the trial values for the facility locations from (P4a) as $y_i^\delta(T)$, and the trial values of resource allocations and the second-stage approximation from (P4b) as $r_i^\delta(T)$ and $\theta^\delta(T)$, respectively. Set the lower bound equal to the sum of the objective values of the optimal solutions of (P4a) and (P4b): $lowerB = \phi_{P4a} + \phi_{P4b}$.

Note that if $T = 0$ (i.e., no constraints of the type specified by Eq. (29) have yet been created), then $\theta^\delta(T) = 0$.

Step 2. Establish an upper bound based on feasibility of constraints (15). If necessary, adjust values for $y_{il}^\delta(T)$ to provide a feasible solution $\bar{y}_{il}^\delta(T)$ by placing a facility large enough to accommodate the resource solution $r_i^\delta(T)$ obtained from (P4b) at each location i . This assumes that there is at least one category of facility size, l , that is large enough to accommodate any potential resource solution. This may be an artificial facility with a very large capacity and a very large fixed cost, so that it will never appear in the final optimal solution, but it ensures that a feasible solution can always be constructed. If the objective value of this feasible solution in (P4a), added to the objective value of (P4b), is less than the current upper bound, ϕ^* , update the upper bound: $upperB = \min\{\phi^*, \phi_{P4a}[\bar{y}^\delta(T)] + \phi_{P4b}\}$.

Step 3. Calculate the error term: $\varepsilon = \frac{upperB - lowerB}{upperB}$.

If ε is less than a predetermined threshold value, then the Lagrangian Relaxation has converged to an approximate solution of (P3). Set the solution values as:

$$y_{il}(T) = \bar{y}_{il}^\delta(T) \quad i \in I, l \in L$$

$$r_i^k(T) = r_i^{k\delta}(T) \quad i \in I, k \in K$$

$$\theta(T) = \theta^\delta(T)$$

and go to Step 5. If ε is larger than the threshold, go to Step 4.

Step 4. Update the Lagrangian multipliers using a sub-gradient:

$$\mu_i^{\delta+1} = \mu_i^\delta + \frac{1}{\delta} \left[\sum_{k \in K} b^k r_i^{k\delta}(T) - \sum_{l \in L} M_l y_{il}^\delta(T) \right]$$

Then return to Step 1.

Step 5. Check the value of the current solution of (P3) obtained from steps 1 through 4. Compute $H[y(T), r(T)]$ and

$$\phi^* = \sum_{i \in I} \sum_{l \in L} F_{il} y_{il}(T) + \sum_{i \in I} \sum_{k \in K} q^k r_i^k(T) + H[y(T), r(T)].$$

If $H[y(T), r(T)] = \theta(T)$ then optimality has been reached. Set $y^* = y(T)$, $r^* = r(T)$, and stop. Otherwise, impose an additional optimality cut. Set $T \leftarrow T + 1$ and compute e_{TT}^k and E_T using Eqs. (12) and (13). Set $\delta = 0$ and $\mu_i^\delta = 0$ for $i \in I$. Then return to Step 1.

This algorithm contains an “inner loop” (steps 1–4) that solves problem (P3) and an “outer loop” (step 5) that solves problem (P2). The algorithm has been coded in C++ to run on a desktop computer.

Section 4 contains a case study to illustrate an application of the model. In Section 5 we offer some indications of the computational efficiency of the LLSM algorithm, based on experiments with the case study data.

4. Case study

A case study focusing on preparedness for hurricane threats in the southeastern US serves to illustrate the Stochastic Mixed Integer Program (SMIP) and to verify the Lagrangian L-shaped Method algorithm. In order to verify the methodology, the case study is small enough to also be solved in extensive form with a commercial software package, but detailed enough to be of interest as an illustration. The network created for the study is composed of 30 nodes and 58 links, as shown in Fig. 2.

Three emergency commodities are considered: water, food and medical kits. Water is assumed to be in units of 1000 gallons. Food is in the form of meals-ready-to-eat (MREs) and is considered in units of 1000 meals. Medical kits designed for emergencies are considered in single units. Costs and space estimates for acquiring, storing and transporting these commodities are designed to be relatively realistic, but should be regarded as illustrative only. Table 1 summarizes the key parameters of these commodities as assumed for this case study. The unmet demand penalty costs for each commodity are assumed to be ten times the resource purchase price, and the holding costs are assumed to be 25% of the purchase price. These values are useful for illustration purposes, but may not match values that might be estimated by ultimate users of the model.

Three possible sizes of facilities are considered, with parameters as shown in Table 2. A facility of any size can be opened at any node in the network.

The Atlantic Oceanographic and Meteorological Laboratory (AOML), a National Oceanic and Atmospheric Administration (NOAA) research facility, estimates that an average of 5.9 hurricanes strike the Atlantic Basin (East and Gulf Coast) states each year, with an average of 2.2 being major hurricanes (AOML, 2006). Historical records from a sample of fifteen hurricanes – ten major (categories 3–5) and five minor (categories 1 and 2) – form the basis for constructing scenarios to represent potential demands and damage in the network. Table 3 lists characteristics of the historical storms that were used to construct the case study demands for water, food and medical kits, and specify the network damage.



Fig. 2. Case study network.

Table 1

Unit purchase price, storage volume occupied and transportation costs for commodities.

Commodity	q^k (\$/unit)	b^k (ft ³ /unit)	Transport cost (\$/unit-mile)
Water (1000 gals)	647.7	144.6	0.3
Food (1000 MREs)	5420	83.33	0.04
Medical kits	140	1.16	5.80E–04

Table 2

Categories, fixed costs and storage capacity of facilities.

Size category	Descriptor	F_i (\$)	M_i (ft ³)
1	Small	19,600	36,400
2	Medium	188,400	408,200
3	Large	300,000	780,000

A set of 51 scenarios represents varying demands and network damage in the case study. The scenarios include both single storms and combinations of two nearly simultaneous storms, as summarized in Table 4. The probabilities assigned to the scenarios are based on approximately matching aggregate historical characteristics of hurricanes in the region, but should be treated as simply illustrative values.

Table 3

Characteristics of sample hurricanes.

Hurricane	Category	Landfall node	Links unusable	Water demand (1000 gallons)	Food demand (1000 units)	Medicine demand (units)
1	3	5	(4, 5)	350	525	500
2	5	14	(12, 14) (14, 15) (15, 24)	560	927	883
3	2	22	–	861	181	402
4	2	22	(17, 20)	9000	1692	3760
5	4	11, 29	–	7500	1771	1687
6	3	15	–	1000	1838	1751
7	2	21	(21, 22)	600	324	720
8	1	11	(8, 12)	1500	162	360
9	5	13, 29	(12, 13)	1040	13,300	95,000
10	2	–	–	2250	1125	18,750
11	3	21	(21, 22)	5000	1750	12,500
12	3	–	(15, 24)	18,000	630	4500
13	3	–	–	2818	80	571
14	4	14, 30	–	2239	1477	10,551
15	4	22	–	4400	3921	28,007

Table 4

Scenario definitions and probability of occurrence.

Single hurricane threats			Combined hurricane threats			
Scenario	Hurricane	Probability	Scenario	Hurricanes	Probability	
1	1	0.02308	16	1	2	0.0046
2	5	0.05	17	1	4	0.0057
3	10	0.16167	18	1	7	0.0057
4	3	0.05363	19	10	2	0.006
5	2	0.00925	20	10	13	0.0261
6	12	0.03083	21	10	9	0.0125
7	13	0.1338	22	10	8	0.0094
8	4	0.05363	23	2	5	0.005
9	11	0.02295	24	2	6	0.0047
10	14	0.02295	25	12	1	0.0052
11	15	0.02295	26	12	3	0.0061
12	7	0.05363	27	12	2	0.0047
13	9	0.05	28	12	4	0.0061
14	8	0.0308	29	12	14	0.0052
15	6	0.03083	30	13	2	0.0057
		Total = 0.75	31	13	8	0.0086
			32	4	2	0.005
			33	11	5	0.0056
			34	11	12	0.0052
			35	11	13	0.0075
			36	11	7	0.0057
			37	14	3	0.0057
			38	14	6	0.0052
			39	15	5	0.0056
			40	15	7	0.0057
			41	15	13	0.0075
			42	15	14	0.005
			43	9	1	0.0056
			44	9	14	0.0056
			45	8	5	0.006
			46	8	3	0.0061
			47	8	7	0.0061
			48	6	5	0.006
			49	6	3	0.0061
			50	6	7	0.0061
			51	No hurricane		0.0174
						Total = 0.25

In the scenario definitions, any facilities (and all supplies stocked there) located at the landfall points for major hurricanes are considered to be destroyed. For minor hurricanes, the facilities and available supplies (if any) at the landfall point are assumed to be damaged (50% loss).

Table 5 summarizes the solution from the LLSM algorithm. Five storage facilities are opened, distributed widely across the network. Four of the five facilities are specialized – storing only one of the three commodity types – and only one facility stores all three commodities. In total, approximately 8.6 million gallons of water are pre-positioned, along with 3.9 million MRE's and approximately 19,000 medical kits. The overall objective function value for this solution is \$90.1 million, with \$30.5 million of this in pre-positioning costs, approximately \$5.3 million in expected transportation and inventory costs, and \$54.3 million in expected penalty costs for unmet demand. Of the pre-positioning costs, the fixed costs make up only about \$1 million of the total, with \$29.5 million for acquisition of material to be stocked.

The large portion of the total expected cost that is penalty for unmet demand is a direct result of the distribution of demands across scenarios. Figs. 3–5 show the cumulative distribution functions of demand for the three commodities, as defined by the 51 scenarios. For all three commodities, the demand distribution has an extreme right-hand tail. For example, the median demand for food is approximately 1 million MREs, but there is about an 8% chance that demand will exceed 4 million MREs, and in the most severe case the demand is nearly 15 million MREs.

Table 5

Solution for nominal version of case study.

City	Facility size	Water (1000 gals)	Food (1000 meals)	Medical kits
Brownsville, TX	Large	2879	0	0
Memphis, TN	Small	0	0	7951
Columbia, NC	Large	2129	3932	8533
Charleston, SC	Small	0	0	2672
Tampa, FL	Large	3596	0	0
	Total =	8604	3932	19,156

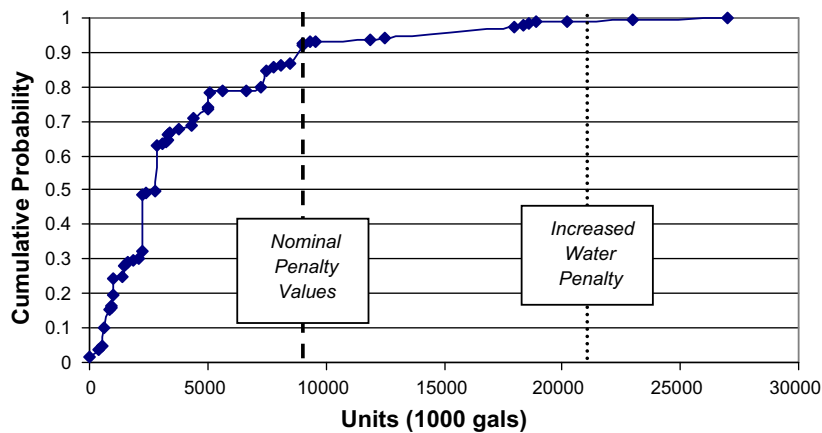


Fig. 3. Cumulative distribution of demand for water and pre-positioned amounts with varying penalties for unmet demand.

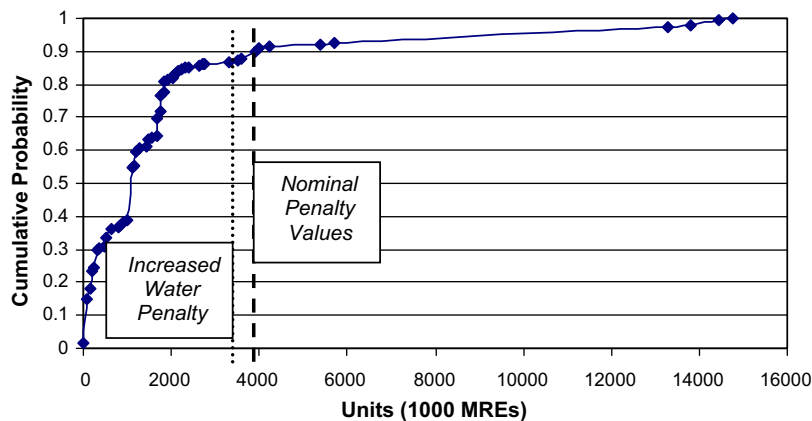


Fig. 4. Cumulative distribution of demand for food and pre-positioned amounts, with varying penalties on unmet demand for water.

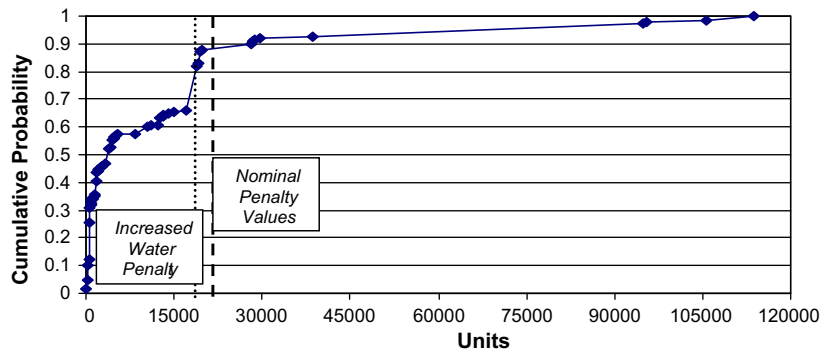


Fig. 5. Cumulative distribution of demand for medical kits and pre-positioned amounts, with varying penalties on unmet demand for water.

Table 6

Solution with increased penalty for unmet water demand.

City	Facility size	Water (1000 gals)	Food (1000 meals)	Medical kits
Brownsville, TX	Medium	0	2001	22
Jackson, MS	Small	0	0	4730
Mobile, AL	Large	5394	0	0
Nashville, TN	Medium	1534	0	0
Columbia, NC	Large	4604	0	6060
Charlotte, NC	Small	0	0	858
Charleston, SC	Medium	2046	0	3069
Tallahassee, FL	Large	4358	0	1991
Tampa, FL	Large	4378	1764	0
Miami, FL	Small	0	0	1574
Total =		22,314	3765	18,304

Figs. 3–5 also show the optimal solution for the total amount of water (8604 units), food (3932 units) and medical supplies (19,156 units) pre-positioned in the network (with the legend “Nominal Penalty Values” indicating the penalty cost value for unmet demand). For food, the solution is to stock material sufficient to cover the demand approximately 90% of the time. For water, the solution stocks material sufficient to cover the demand 86% of the time; and for medical kits, 83% of the time. The solution does not stock sufficient material to cover all the demand in the worst-case scenarios because those scenarios occur with small probability. The model is balancing the penalty costs for unmet demand in a few scenarios against the costs of having unused material in most of the cases.

As an indication of the sensitivity of the solution to input parameter values, a second experiment is shown, where water is given greater importance (the penalty cost for lack of water is increased by a factor of 10, to one hundred times its purchase price). The unmet demand penalty costs for food and medicine are kept at ten times their purchase price. The resulting solution for the amount of pre-positioned material is shown in Figs. 3–5 with the legend “Increased Water Penalty”. Table 6 summarizes the solution.

In this case, the solution has a total of 10 open facilities, six of which store water. Four of the facility locations are the same as in the previous experiment, but the sizes and mix of material stocked are different. As shown in Figs. 3–5, the solution allocates a total amount of 22,314 units of water, 3765 units of food and 18,304 units of medical supplies. The amount of water stocked is sufficient to cover demand approximately 99% of the time. The increase in penalty costs for water results in small decreases in the amount of food and medicine pre-positioned. For food, the solution allocates enough stock to meet the demand approximately 88% of the time, a decrease from 90% in the previous experiment. For medical kits, demand is met approximately 80% of the time instead of 83%. The overall expected cost is \$99.0 million, composed of \$39.2 million in pre-positioning costs, \$6.9 million in transportation costs, and \$52.9 in expected unmet demand penalties. The overall expected cost increases about 10% from the previous experiment, including a substantial increase in pre-positioning costs (29%) and transportation costs (30%), partially offset by a reduction in expected penalty costs (3%).

When some highway links are unusable, transportation costs increase because some shipments of supplies must use more circuitous routes. It is also possible in some scenario definitions to have parts of the network completely cut off as a result of unusable links (although this does not occur in the scenarios defined for the current case study). To overcome large-scale network damage, supplies can be moved by air. This can be represented in the model by creating additional links. These links provide an expensive alternative, but one that is not susceptible to damage by specific storms. Capacity limits can also be added on these links to reflect limited ability to move materials, if desired. For example, Fig. 6 shows addition of air links servicing areas prone to major hurricane damage (i.e., Wilmington, Charleston, New Orleans, Baton Rouge, Biloxi and

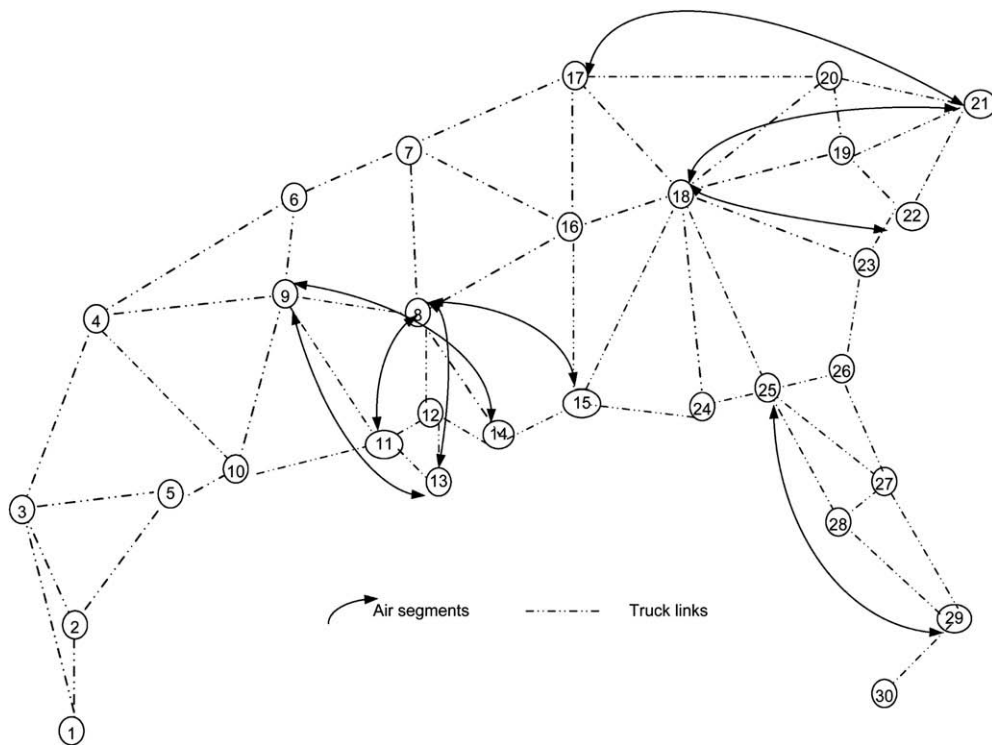


Fig. 6. Air links added to the network.

Miami). These links can be useful to prevent demand areas from being cut off from the network in the case of extensive flooding, with resulting large penalty costs for unmet demand.

5. Evaluation of solution quality and computational speed

For the nominal data input in the case study, the LLSM algorithm achieved a solution (summarized in Table 5) with an objective value of \$90.1 million. This solution was obtained in 51 s on a PC with a 2.80 Ghz Pentium 4 processor and 512 MB of memory. To evaluate the quality of this solution, we have also solved the problem in extensive form using CPLEX. This produced an optimal solution with an objective function value of \$89.5 million (approximately 0.6% better) in 685 s on the same computer.

The facility locations and pre-positioning quantities from the CPLEX solution are summarized in Table 7. The CPLEX solution opens a greater number of facilities compared to the LLSM solution, but it uses mostly small facilities (only one medium and one large facility). The total first-stage cost in the CPLEX solution (facilities and material acquisition) is \$28.9 million, which is slightly lower than the cost in the LLSM solution. This is primarily the result of opening a greater number of small facilities instead of few large (more expensive) facilities. However, the second-stage cost in the CPLEX solution is slightly higher than in the LLSM solution. Overall, although the facility locations chosen are different, these two solutions perform similarly considering total expected cost and percentage of satisfied demand. The CPLEX solution meets water demand 92% of the time, food 88%, and medicine 87%.

Across a somewhat larger set of test problems (with varying demands, network configurations, unmet demand penalties, etc.), we have found that the LLSM algorithm consistently achieves a solution within 0.7% of optimality with a running time that is between 7% and 31% of that required by CPLEX.

One of the advantages of the case study described in Section 4 is that it is large enough to be realistic, but still small enough to allow extensive solution using a commercial MIP solver to allow evaluation of the solution quality and computing time of the LLSM algorithm. This evaluation has provided confirmation of the solution quality produced by the LLSM approach and also indicates a computational advantage for the LLSM algorithm. On larger problems (more commodities, a more detailed network, more scenarios, or some combination of the three), the LLSM approach will scale relatively easily because it decomposes the problem. Extensive solution using a commercial MIP solver, however, fairly quickly becomes prohibitive.

For example, in the case study problem there are three commodities, 51 scenarios and the network has 30 nodes and 58 links. The extensive form MIP has 18,234 variables, 90 of which are binary, and 4650 constraints (ignoring link capacity and non-negativity). A modest increase in problem size might involve 10 commodities, 100 scenarios, and a network with 200

Table 7

Solution for nominal version of case study using CPLEX.

City	Facility size	Water (1000 gals)	Food (1000 meals)	Medical kits
Lake Charles, LA	Small	99	253	813
Baton Rouge, LA	Small	252	0	0
New Orleans, LA	Small	252	0	0
Mobile, AL	Small	219	0	4069
Atlanta, GA	Small	252	0	0
Columbia, NC	Large	4184	1984	8376
Charlotte, NC	Small	252	0	0
Charleston, SC	Small	0	0	2373
Savannah, GA	Small	252	0	0
Tallahassee, FL	Medium	2000	1372	4000
Int. 110 and 175	Small	252	0	0
Jacksonville, FL	Small	252	0	0
Orlando, FL	Small	252	0	0
Tampa, FL	Small	252	0	0
Key West, FL	Small	231	0	0
Total =		9001	3609	19,631

nodes and 600 arcs. The extensive form for this problem would have more than 1 million variables (600 of which are binary) and about 200,000 constraints. Solution of such a problem is likely to be possible with a good commercial MIP solver like CPLEX, but would be very time consuming. Using the LLSM approach on this expanded problem requires solving for minimum cost flows on networks of roughly 400 nodes and 1000 arcs, which is not difficult at all. The number of these sub-problems increases from 150 at a given iteration to 1000, but this is also a relatively modest increase. The LP to be solved in step 1 of the algorithm [problem (P4b)] has 2000 variables and a relatively small number of constraints (perhaps up to a few hundred, based on the number of iterations), and this is still a very small LP problem. The relative advantages of the LLSM algorithm are more and more pronounced as the problem size grows.

6. Conclusions and future work

Pre-positioning of emergency supplies can be an effective mechanism for improving response to natural disasters. We have created an optimization model whose solution provides a pre-positioning strategy for facility locations and resource stocking under uncertainty about the eventual demands to be faced. The model is a stochastic mixed-integer programming problem. For relatively small data instances, the problem can be solved in extensive form using a commercial mixed-integer solver, but this solution approach does not scale well to large problem instances. For this reason, a heuristic solution algorithm was devised to exploit the embedded network structure of the problem. This solution combines the integer L-shaped method for addressing stochastic programming problems and Lagrangian relaxation for solving one of the embedded sub-problems.

The resulting algorithm, referred to as the Lagrangian L-shaped method (LLSM), solves the overall stochastic mixed-integer problem by decomposing it into a series of smaller and more easily solved sub-problems. This approach scales well, to allow very large problems to be solved.

A case study addressing hurricane risks in the southeastern US illustrates both the formulation and the character of the problem solutions in a practical context. This case study has also allowed experiments to assess the solution quality and computational effectiveness of the LLSM algorithm. Across a series of test problems, the LLSM algorithm consistently produces solutions whose objective function values are within 0.7% of optimal, with computation times that are between 7% and 31% of the time required using commercial MIP software. Based on the experiments, we are confident that the LLSM can be used as a large-scale resource pre-positioning planning tool.

In the test cases analyzed here, there are multiple solutions that have nearly the same total expected cost. The solutions found by the LLSM algorithm and CPLEX are often somewhat different (in number of facilities and acquisition of resources), but they have very similar objective values. This characteristic of the objective function provides flexibility in the selection of locations for operating facilities. The model may provide an array of alternatives in the selection of facility locations without large cost differences, thus permitting government officials the opportunity of considering other elements that may not be quantifiable (e.g., political concerns, availability of facility locations).

Several avenues present themselves as directions for further work. A key assumption has been made in developing the LLSM algorithm that the important character of link capacity constraints for transportation of emergency supplies is whether the link is usable or not. If it is, it is assumed to have sufficient capacity to carry whatever quantity of supplies is required. This is likely to be a useful representation for many situations, but it is clearly a potential limitation of the model. A generalization that treats the multi-commodity flow problem in the second stage directly is certainly worth developing. Such a generalization loses one of the attractive computational elements of the LLSM algorithm – the ability to decompose the network flow problems by commodity – but would increase the range of potential applicability for the model as a whole.

The character of the probability distributions for demand seen in the case study also indicates a desirable direction for further thought about the objective function in the model. On its face, minimizing total expected cost is a reasonable objective, but when there are a few scenarios that have extreme demands, computing expected costs may tend to hide underlying issues rather than illuminate them. Planning decisions driven solely by worst-case scenarios are seldom affordable, but some “middle ground” between expected values and worst-case planning is likely to be desirable. Daskin et al. (1997) explored this idea in the context of p -median location problems under uncertain demand, and that may prove to be a productive direction for the pre-positioning application as well.

References

- Ahuja, R.K., Magnanti, T.L., Orlin, J.B., 1993. *Network Flows: Theory, Algorithms, and Applications*. Prentice Hall, New Jersey.
- Al-qurashi, F., 2004. New vision of emergency response planning. *Process Safety Progress* 23 (1), 56–61.
- Atlantic Oceanographic and Meteorological Laboratory, 2006. Frequently asked questions: E11) How many tropical cyclones have there been each year in the Atlantic basin? What years were the greatest and fewest seen? National Oceanic and Atmospheric Administration. <<http://www.aoml.noaa.gov/hrd/tcfaq/E11.html>>.
- Balcik, B., Beamon, B.M., 2008. Facility location in humanitarian relief. *International Journal of Logistics: Research and Applications* 11 (2), 101–121.
- Barbarosoglu, G., Arda, Y., 2004. A two-stage stochastic programming framework for transportation planning in disaster response. *The Journal of the Operational Research Society* 55 (1), 43.
- Birge, J.R., Louveaux, F., 1997. *Introduction to Stochastic Programming*. Springer-Verlag, New York.
- Daskin, M.S., Hesse, S.M., Reville, C.S., 1997. α -Reliable p -minimax regret: a new model for strategic facility location modeling. *Location Science* 5 (4), 227–246.
- Fiedrich, F., Gehbauer, F., Rickers, U., 2000. Optimized resource allocation for emergency response after earthquake disasters. *Safety Science* 35 (1), 41–57.
- Fisher, M.L., 1985. An applications oriented guide to lagrangian relaxation. *Interfaces* 15 (2), 10–21.
- Haghani, A., Oh, S.-C., 1996. Formulation and solution of a multi-commodity, multi-modal network flow model for disaster relief operations. *Transportation Research Part A: Policy and Practice* 30 (3), 231–250.
- Houming, F., Tong, Z., Xiaoyan, Z., Mingbao, J., Guosong, D., 2008. Research on emergency relief goods distribution after regional natural disaster occurring. In: *International Conference on Information Management and Industrial Engineering*.
- Johnstone, D.P., Hill, R.R., Moore, J.T., 2004. Mathematically modeling munitions prepositioning and movement. *Mathematical and Computer Modelling* 39 (6–8), 759–772.
- Laporte, G., Louveaux, F.V., 1993. The integer L-shaped method for stochastic integer programs with complete recourse. *Operations Research Letters* 13 (3), 133–142.
- Ozdamar, L., Ekinici, E., Kucukyazici, B., 2004. Emergency logistics planning in natural disasters. *Annals of Operations Research* 129 (1–4), 217–245.
- Parentela, E., Nambisan, S.S., 2000. Emergency response (disaster management). *Urban Planning and Development Applications of GIS*, 181–196.
- Saccommanno, F.F., Allen, B., 1988. Locating emergency response capability for dangerous goods incidents on a road network. *Transportation Research Record* 1193, 1–9.
- Sathe, A., Miller-Hooks, E., 2005. Optimizing location and relocation of response units in guarding critical facilities. *Transportation Research Record* 1923, 127–136.
- Sherali, H.D., Desai, J., Glickman, T.S., 2004. Allocating emergency response resources to minimize risk with equity considerations. *American Journal of Mathematical and Management Sciences* 24 (3–4), 367–410.
- Sheu, J.-B., 2007. An emergency logistics distribution approach for quick response to urgent relief demand in disasters. *Transportation Research Part E: Logistics and Transportation Review* 43 (6), 687.
- Slyke, R.M.V., Wets, R., 1969. L-shaped linear programs with applications to optimal control and stochastic programming. *Journal on Applied Mathematics* 17 (4), 638–663.
- Tsai, Y., Wang, Z., Yang, C., 2002. A prototype real-time GPS/GIS-based emergency response system for locating and dispatching moving patrol vehicles with the beat-based shortest distance search. *Traffic and Transportation Studies Proceedings of ICTTS 2*, 1361–1368.
- Ukkusuri, S., Yushimito, W., 2008. Location routing approach for the humanitarian prepositioning problem. *Transportation Research Record* 2089, 18–25.